LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	18
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Capacitance type fuel gauge
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Capacitance type fuel gauge.
5. Specific Objectives	15.1.7.3 Trainee will be able to explain Capacitance type fuel quantity measurement indicator.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Capacitance type fuel gauge theory diagrams and circuit operation principles.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the basic definition of capacitance?
(b) How does a fuel quantity indicator generally work?
(c) What are the two main types of fuel gauges used in aircraft?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principle of operation based on dielectric constant • Structure of probe and capacitor plates • Effect of fuel level on capacitance value • Signal conditioning and indicator display	L & Ds	PPT, IB	15 min
(b)	• Advantages of capacitance type over other types • Limitations regarding fuel type and contamination • Maintenance requirements and inspection points • Troubleshooting common failure modes	L & Ds	PPT, PDF	20 min
(c)	• Interpretation of gauge readings and calibration • Impact of temperature on measurement accuracy • Safety considerations during system checks • Integration with aircraft fuel management system	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the dielectric constant changes with fuel level.
(b) Describe one major limitation of the capacitance type fuel gauge.
(c) List two safety precautions when inspecting the fuel probe.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between capacitance type and resistive type fuel gauges to be discussed in the next lesson on Synchroscope introduction.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to Synchroscope	Refer to the master syllabus reference list	LP 19

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Wing Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min